

Nathaniel Trueba

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EDUCATION

University of California, San Diego

Bachelor of Science in Data Science and Machine Learning

San Diego, CA

September 2025 – May 2027

WORK EXPERIENCE

Undergraduate Student Assistant – Lenain Air-Sea Interaction Lab

December 2025 – Present

University of California, San Diego

San Diego, CA

- Engineered **research-grade digital elevation and surface models** from LiDAR datasets, validated through systematic quality-control checks and complete metadata verification for coastal terrain analysis.
- Classified** and manually refined **LiDAR point clouds** to extract control points and derive shoreline contours.
- Built **GIS/LiDAR** workflows using **QGIS** and **point-cloud** tools for terrain modeling and flood-risk assessment.
- Designed and maintained **structured geospatial data pipelines** and documentation standards, ensuring reproducibility and long-term usability of research outputs across multiple GIS and modeling-assessment projects.

PROJECTS

TubeScope - Lifecycle Analytics of Trending YouTube Videos | *Python, YouTube API, Pandas, Streamlit*

- Collaborated with a team to design and implement an **end-to-end analytics pipeline** studying YouTube virality, aligning contributions across data collection, modeling, and visualization.
- Automated collection** and analysis of daily trending data to study **viral content longevity** for 1000+ videos.
- Applied **Kaplan-Meier survival analysis** to measure video “half-lives” and identify critical features.
- Designed a predictive model using **linear regression** and **interactive dashboard** to forecast viral lifespan.
- Built and deployed a Streamlit dashboard that visualize virality trajectory, survival curves and predicted duration.

Surf Quality Forecast Dashboard | *Python, Pandas, NumPy, Matplotlib, Time Series Analysis, API Integration*

- Built an end-to-end forecast dashboard with real-time buoy, tide and wind data to create a **Surf Quality Index**.
- Engineered **domain-specific features** including wave steepness, swell-coastline alignment, wind direction flags, and tide band scoring to improve predictive accuracy with **automated detection** of 2 hour “best surf windows”.
- Designed **user-friendly visualizations** and outputs, enabling users to easily interpret forecasted conditions.
- Deployed a **live notification system** to send SMS alerts when predicted conditions cross **user-defined thresholds**, showcasing applied data engineering of **NOAA time-series** data and **UX design**.

LEADERSHIP AND MANAGEMENT

Computer Science Club – Vice President of Data Science

September 2024 – June 2025

San Diego Miramar College

San Diego, CA

- Revitalized the organization by expanding resources, industry access, and student opportunities for club members.
- Designed, presented and tutored guided student projects, **mentoring peers** in data science concepts to foster hands-on learning and build real-world technical and soft skills for beginners and experienced students alike.
- Engaged industry professionals through networking and hosted talks and events, providing club members with exposure to real-world applications, relevant information about early career timelines and various career pathways.

SKILLS

Programming & Visualization: Pandas, SQL, NumPy, SciKit-Learn, Plotly, Tableau, Matplotlib, Seaborn, AWS

Machine Learning & Predictive Modeling: Regression, Feature Engineering, Time-Series Analysis, Ensemble methods (XGBoost, Random Forest), ARIMA, Classification & Clustering (K-Means), Neural Networks (MLP), Natural Language Processing

Statistics and Mathematics: Hypothesis testing, A/B testing, Probability, Descriptive statistics, Linear algebra, Optimization, Multivariate analysis, Principal Component Analysis

Relevant Coursework: Python, Data Structures & Algorithms, Java, C++, Linear Algebra, Data Science, Exploratory Data Analysis, Theoretical Foundations of Machine Learning, Statistical Methods